



Hardware-Mapped Neural Inference Engine for Real-Time Elderly Fall Detection

SDG-9 Industry Innovation and Infrastructure

Team Introduction

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Presentation Outline

- Introduction
- Literature Review
- Problem Definition
- Objectives
- Methodology
- References

Introduction

- Elderly falls are a major cause of accidental injuries, creating the need for reliable real-time monitoring systems.
- Many existing fall detection systems rely on software processing or cloud-based AI, increasing latency and power consumption.
- This project proposes COGXAR, a hardware-based neural inference system for real-time fall detection.
- The system uses multi-sensor fusion involving motion, pressure, and sound sensing along with hardware acceleration on the PSoC platform to enable efficient and low-cost detection.

Literature Review

Title	Author	Publication	Abstract
A Smart Wearable-Based Fall Detection and Health Monitoring System for Elderly Care Using IoT and Machine Learning	G. Santhanamari et al.	IEEE / 2025 International Conference on Next Generation Computing Systems (ICNGCS) (2025)	This paper presents a wearable system using MPU6050 and MAX30100 sensors with a Random Forest classifier (97% accuracy) for real-time fall detection and health monitoring, with GPS/GSM-based emergency alerts and cloud dashboard visualization.
Embedded Machine Learning Using Microcontrollers in Wearable and Ambulatory Systems for Health and Care Applications: A Review	M. S. Diab, E. Rodriguez-Villegas	IEEE Access (2022)	This review investigates the use of microcontroller units (MCUs) as edge inference devices for TinyML-based wearable healthcare systems, covering hardware specifications, software frameworks, and existing health and care applications including fall detection.
Unveiling Fall Origins: Leveraging Wearable Sensors to Detect Pre-Impact Fall Causes	S. Kiran et al.	IEEE Sensors Journal (2024)	This paper proposes a CNN-BiGRU deep classifier for pre-impact fall detection using wearable IMU sensors, achieving 98% accuracy across 36 activities (15 fall types, 21 ADLs) from the KFall dataset with an inference time of 46–52 ms

Literature Review

Title	Author	Publication	Abstract
Edge AI-Based Fall Detection for an Elderly Care Robot	Joseph Haggege et al.	IEEE / Research Publication (2026)	This paper presents a lightweight edge AI fall detection system deployed on NVIDIA Jetson Nano for real-time elderly monitoring with privacy-preserving on-device inference.
A Lightweight AI System for Real-Time Elderly Fall Detection	M. Fayad et al.	Springer Nature / Computing Journal (2026)	This paper presents a lightweight AI-based embedded system for accurate elderly fall detection using real-time sensor fusion and efficient inference techniques.
Artificial Intelligence for Fall Detection in Older Adults	A. Gattani et al.	IEEE / Artificial Intelligence in Medicine (2025)	This paper reviews AI-based fall detection systems using wearable sensors and embedded intelligence for continuous real-time elderly monitoring.
Privacy-Preserving Fall Detection at the Edge Using Neuromorphic Processor	Lyes Khacef et al.	Neuromorphic Edge AI Research (2025)	This work proposes a low-power edge AI fall detection system using neuromorphic processors and event-based sensors for real-time healthcare monitoring.

Research Gaps

- **Software-Based AI Dependency:** Most existing fall detection systems rely on software execution, cloud processing, or external GPUs, resulting in increased latency and power consumption.
- **Lack of Hardware-Level Neural Inference:** Current embedded AI systems mainly execute neural network computations on microcontroller cores rather than programmable hardware resources.
- **Limited Multi-Sensor Fusion:** Many traditional fall detection systems use only accelerometer data, reducing classification accuracy in real-world scenarios.
- **Need for Low-Cost Real-Time Edge AI:** There is a growing requirement for compact, low-power, and low-cost embedded AI systems capable of real-time healthcare monitoring without cloud dependency.

Problem Definition

“To address the challenge of high latency and cloud dependency in existing software-based elderly fall detection systems.”

Objectives

- To develop a real-time elderly fall detection system using multi-sensor fusion.
- To implement neural network multiply-accumulate (MAC) operations.
- To achieve low-cost, low-latency, and cloud-independent embedded edge-based fall detection.



Methodology

1. Data Acquisition

Collect real-time data from sensors using I²C interface.



MPU6050
(Accelerometer + Gyroscope)



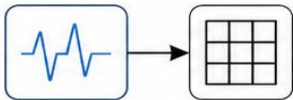
BMP280
(Pressure + Temperature)



KY-038
Sound Sensor
(Sound Level)

2. Data Preprocessing

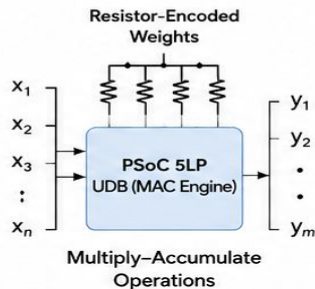
Filter, normalize and extract relevant features.



- Noise Filtering
- Normalization
- Feature Vector Formation

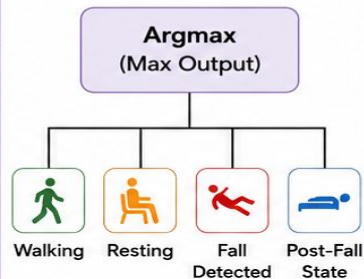
3. Hardware-based Neural Inference

Perform MAC operations using PSoC 5LP UDBs with resistor-encoded weights.



4. Activity Classification

Use argmax logic to identify the activity.



5. Output & Indication

Display results and alert the user.



LED Indication



UART Output

System Platform

Sensors

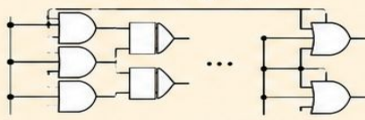
- MPU6050
- BMP280
- KY-038 Sound Sensor

I²C

PSoC 5LP (Controller)



UDB Datapath (MAC Engine)



Classifier (Argmax)



Outputs

- LCD
- LED
- UART

Methodology

1 Sensor Acquisition

Three I²C sensors provide raw environmental inputs:



- **MPU6050** – Acceleration magnitude (primary fall indicator)



- **BMP280** – Pressure / posture context



- **KY-038** – Sound level (impact / activity proxy)

The PSoC hardware **I²C master** reads these sensor values.

$$X = [x_{accel}, x_{press}, x_{sound}]$$

2 Sensor Normalization

Sensor readings have different ranges, so they are scaled to **[0,1]** using min-max normalization.

$$x_{norm} = \frac{x_{raw} - x_{min}}{x_{max} - x_{min}}$$

Normalized input vector:

$$x_{norm} = [x_0, x_1, x_2]$$

Methodology

3. Neural Weight Matrix

A 4×3 weight matrix represents the neural model.

- **4 neurons** $\rightarrow$ 4 activity classes
- **3 inputs** $\rightarrow$ 3 sensor signals

$$W = \begin{bmatrix} W_{00} & W_{01} & W_{02} \\ W_{10} & W_{11} & W_{12} \\ W_{20} & W_{21} & W_{22} \\ W_{30} & W_{31} & W_{32} \end{bmatrix}$$

Bias for each neuron:

$$B = [B_0, B_1, B_2, B_3]$$

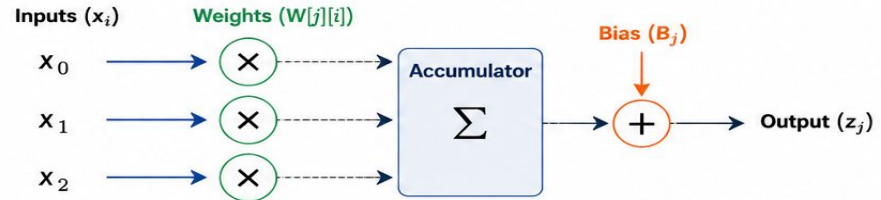
Each neuron also includes a bias B_j .

4. Hardware MAC Engine (UDB)

Four **PSoC Universal Digital Block (UDB)** datapaths perform the multiply-accumulate operation in hardware.

For each neuron:

$$z_j = \sum_{i=0}^2 W[j][i] \cdot x_i + B_j$$



The ARM core writes **weights and inputs** into **UDB registers**, and the **UDB datapath** performs the multiplication and accumulation.

Methodology

5. Output Interface

The detected class is displayed through:

- **UART** terminal output
- **LED** alert for fall detection

UART Terminal Output



LED Alert for Fall Detection



LED Alert



Results

```
Sensors | A=86 P=98000 S=740 | Norm[147  0 12]
HW=448 cyc | SW=1016 cyc | [CLASS] RESTING
Sensors | A=29 P=98000 S=745 | Norm[ 50  0 14]
HW=447 cyc | SW=1016 cyc | [CLASS] FALL DETECTED
Sensors | A=1383 P=98000 S=741 | Norm[255  0 13]
HW=448 cyc | SW=1016 cyc | [CLASS] RESTING
Sensors | A=1439 P=98000 S=741 | Norm[255  0 13]
HW=448 cyc | SW=1016 cyc | [CLASS] RESTING
Sensors | A=24 P=98000 S=747 | Norm[ 40  0 14]
HW=448 cyc | SW=1017 cyc | [CLASS] NORMAL WALKING
```

Sensor Data and
Classification

*Go, change the
world*

Expected Outcomes

- To show that the UDB block is faster for MAC implementation than Software.
- To build an edge based fall detection system.
- To publish a IEEE conference paper.

Conclusion

- **Successful Implementation of Edge-Based Inference:** Developed a fully cloud-independent, low-cost, and real-time elderly fall detection system using a multi-sensor fusion approach (integrating motion, pressure, and sound data) to maximize classification accuracy in practical environments.
- **Hardware Acceleration vs. Software Processing:** Successfully demonstrated that offloading neural network multiply-accumulate (MAC) operations to the PSoC 5LP Universal Digital Blocks (UDB) significantly improves performance, cutting execution time roughly in half (~448 hardware cycles compared to ~1016 software cycles).
- **Fulfillment of Project Goals:** Met all critical design milestones—including data normalization, hardware-mapped matrix computations, and real-time output categorization (Walking, Resting, Fall Detected, Post-Fall State)—establishing a viable blueprint for next-generation, privacy-preserving wearable health monitors.

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Thank You!